PhD Research Proposal: Advanced Numerical Modelling of Terahertz Transmission Lines for Next-Generation Communication Systems

Name: Mohammed Al Shuaili
Supervisor: Dr. Marissa Condon
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Section One: Introduction:

The impending deployment of 6G networks and the exponential growth in demand for wireless data rates exceeding 100 Gbit/s have catalysed research into the terahertz (THz) frequency band (0.1–10 THz) [1]. This spectrum offers vast bandwidths for applications ranging from ultra-fast wireless networking to biomedical imaging. However, a critical bottleneck is the accurate and efficient modelling of signal propagation at these frequencies, where transmission lines (TLs) exhibit complex, frequency-dependent behaviours such as strong dispersion and the skin effect. Conventional time-domain methods (e.g., FDTD, RLC ladders) struggle with these phenomena as they assume time-invariant parameters.

While advanced algorithms such as Asymptotic Waveform Evaluation (AWE) and Vector Fitting (VF) have succeeded in microwave regimes, their performance and optimal implementation for THz TLs are poorly understood. This PhD research proposes a novel, systematic examination of these advanced model-order reduction techniques, with a focus on AWE and its hybridization with other algorithms such as VF and fast convolution**.** The primary objective is not merely incremental improvement but to create and validate a robust, computationally efficient modelling framework that is specifically tailored for the THz spectrum. This framework will accurately incorporate fractional-order, frequency-dependent losses, enabling the optimised design of high-speed THz interconnects and antennas for future communication systems.

Section Two: Preliminary Literature Review

The literature on high-frequency transmission line modelling is vast, evolving from fundamental electromagnetic theory to sophisticated numerical techniques designed to overcome the limitations of traditional circuit simulators as frequencies ascend into the microwave, millimetre-wave, and now, terahertz regimes. The demand for data rates exceeding 100 Gbit/s for 6G and beyond has thrust the terahertz (THz) band (0.1–10 THz) into the research spotlight [1]. This spectrum promises revolutionary applications but is plagued by the "THz gap," where signal propagation is dominated by complex, frequency-dependent phenomena such as strong dispersion and the skin effect. Traditional transmission line (TL) modelling techniques, which often assume time-invariant parameters, are fundamentally ill-suited for this regime.

Recent Advances and the Path to THz:

Critically, recent research has made significant strides in enhancing these techniques, directly informing the proposed work:

- **Innovative Convolution Schemes:** Petranice et al. (2025) **[2]** developed fast, piecewise-linear convolution schemes (PWL-SFC) for large-scale electromagnetic systems. Their "quasi-recursive" formulation offers a substantial computational speed-up while maintaining accuracy, providing a powerful alternative to pole-residue based convolution for time-domain simulation.

- **Advanced NILT for Complex Systems:** Keran and Dounavis (2023) **[3]** demonstrated a highly efficient NILT-based algorithm for coupled RLC interconnects, using modal decoupling to achieve near-linear computational complexity. This proves NILT's viability for large-scale systems relevant to integrated THz circuits. Furthermore, Bandali et al. (2025) **[4]** introduced a modified NILT (NILTn) framework for nonlinear circuits, achieving high-order, stable time-stepping. This is a breakthrough, showing that the high accuracy of Laplace-domain methods can be extended to practical, non-linear circuit environments.

- **Enhanced Vector Fitting for Complex Electromagnetic Parameters:** The ongoing refinement of core Model Order Reduction (MOR) algorithms remains critical for THz applications. Very recently, Li et al. (2025) **[5]** proposed an accuracy-enhanced Vector Fitting (VF) technique that addresses a fundamental limitation of traditional VF: the enforced conjugate symmetry of complex pole-residue pairs. For certain electromagnetic parameters—such as the complex conductivity of graphene or the permittivity of metals at optical/THz frequencies—this symmetry is not physically required and can introduce fitting errors. Their method introduces a set of "gain coefficients" for each term in the pole-residue model, which are solved via a final linear least-squares step. This post-processing adjustment, which leaves the identified poles unchanged, was shown to reduce fitting errors by over 80% in some cases without a significant computational overhead, providing a more accurate rational approximation for materials common in THz devices.

- **Advancements in Stabilized Time-Domain Methods:** Alongside frequency-domain MOR techniques, there is continued development in direct time-domain discretization methods aimed at improving their stability and accuracy. Very recently, Ait Belaid et al. (2024) [6] introduced a novel hybrid algorithm for the transient analysis of lossy, nonuniform multiconductor transmission lines. Their method combines a multistep θ-method in the spatial domain with the Runge-midpoint method for time advancement. This approach was specifically designed to eliminate the numerical oscillations that plague the simple FDTD method, achieving high-order accuracy in both space and time while maintaining stability under the CFL condition. The results demonstrate superior accuracy and significantly reduced oscillations compared to traditional FDTD when simulating complex interconnect structures, highlighting a meaningful step forward for time-domain analysis.

Further validating the potential of the θ-method, a very recent study by Ait errais et al. (2024) [7] specifically applies this technique to the critical challenge of modelling skin effect in high-speed interconnects. Their work develops a formulation that explicitly incorporates a skin impedance term, $Z(s) = R + K\sqrt{s}$, into the telegrapher's equations and solves them using the spatially and temporally discrete θ-method. The results demonstrate a significant advantage over conventional FDTD, achieving a remarkable reduction in simulation time (e.g., 3.65 seconds vs. 50.93 seconds for a test case) while maintaining high accuracy and improved

stability, particularly at higher frequencies corresponding to shorter rise times. This study provides strong evidence that advanced, unconditionally stable time-domain methods such as the θ-method are not only viable but are superior alternatives for efficient and accurate simulation of frequency-dependent losses, a conclusion that directly informs and supports the need for the comparative analysis proposed in this research.

Addressing Frequency Dependence with Fractional-Order Calculus:

The integration of fractional-order calculus to model the skin effect and dispersion more accurately has emerged as a pivotal advancement, moving beyond the simplistic $\sqrt{\omega}$ approximations used in traditional integer-order models [8]. This approach is essential for achieving true physical fidelity at THz frequencies, where the frequency-dependent behaviour of per-unit-length parameters dominates. The foundational work by Shang et al. (2013) [9] demonstrated this efficiently for CMOS on-chip transmission lines. They developed a causal, compact fractional-order RLGC model where the I-V relationships of the inductance and capacitance are redefined using fractional derivatives:

- Fractional-order inductor: $Z_L(s) = s^{\alpha_L} L'$
- Fractional-order capacitor: $Y_C(s) = s^{\alpha_C} C'$

Here, α_L and α_C are the fractional orders (typically between 0 and 1), and L' and C' are pseudo-parameters with non-standard units. This formulation intrinsically generates frequency-dependent loss (dispersion) in the real parts of the impedance and admittance, directly modelling the non-quasi-static effects and complex dispersion at THz bands [9]. A critical contribution of their work was proving that this fractional-order model inherently ensures causality (satisfying the Kramers-Kronig relations) when $\alpha_L + \alpha_C < 2$, a condition routinely violated by integer-order models. Their measurement validated results showed that while the integer-order model deviated above 10 GHz, their fractional-order model accurately predicted the characteristic impedance up to 110 GHz. Furthermore, they demonstrated the profound impact on circuit performance, showing significant differences in signal delay, LO rejection in mixers, and even the oscillation frequency and start-up conditions in standing-wave oscillators.

Building on this, Al-Zubaidi R-Smith et al. (2017) [8] provided a comprehensive methodology for simulating these fractional-order, lossy transmission lines, explicitly incorporating the skin effect. Their model augmented the fractional-order series impedance with a term $K\sqrt{s}$ to represent the high-frequency internal impedance:

$$Z_{0f}(s) = R_0 + s^{\alpha} L'_0 + K\sqrt{s}$$

Their analysis highlighted the critical role of carefully selecting the fractional orders α and β, bounding their values by ensuring the real parts of the characteristic impedance and propagation constant remain physically meaningful (positive for attenuation). Using a sophisticated Numerical Inverse Laplace Transform (NILT) method, they simulated time-domain waveforms and showed that the fractional-order model (e.g., with $\alpha = \beta = 0.98$) captures "abnormal diffusion" phenomena of the voltage wave along the line—a behaviour that

the classical integer model completely neglects. This work underscores that even slight deviations of the fractional orders from unity (1.0) provide a "high degree of flexibility" to more realistically model lossy TLs with frequency dependence**.**

The Shift to Advanced Frequency Dependent Approaches:

The core question the literature addresses is: How can we accurately and efficiently predict the transient and steady-state behaviour of signals propagating on transmission lines, especially when traditional assumptions break down at THz frequencies.

This has catalysed the development of sophisticated frequency-domain methods. The Numerical Inverse Laplace Transform (NILT) provides a benchmark for linear circuits by converting precise s-domain models to the time domain [10]. Model Order Reduction (MOR) techniques such as Asymptotic Waveform Evaluation (AWE) [11] and Vector Fitting (VF) [12] [13] are critical for creating compact, efficient models. AWE uses moment-matching to generate low-order pole-residue models, while VF iteratively fits rational functions to frequency-domain data. The experimental analysis and results of these methods over SPICE for high-speed interconnects is well documented [14] [15].

The Inadequacy of Conventional Methods:

The foundational concept governing TL behaviour is described by the Telegrapher's equations, derived from Maxwell's equations [16]. These coupled partial differential equations relate voltage (v(x,t)) and current (i(x,t)) along the line via per-unit-length parameters: resistance (R), inductance (L), conductance (G), and capacitance (C). A noticeable shift occurs at high frequencies, the per-unit-length parameters (R, L, G, C) become strongly frequency-dependent. The skin effect causes resistance (R) to scale with $\sqrt{\omega}$ [9], while dielectric losses lead to a frequency-dependent conductance (G). Direct time-domain discretisation methods like FDTD [17] become computationally prohibitive due to stringent stability conditions. This dispersion distorts signals and is a primary modelling challenge at THz frequencies [18]. Similarly, lumped RLC ladder approximations [19] suffer from discretisation errors and cannot inherently capture the distributed, frequency-variant nature of THz propagation.

- **Direct Time-Domain Discretisation:** The Finite-Difference Time-Domain (FDTD) method **[18]** discretises the Telegrapher's equations directly on a staggered grid. While conceptually straightforward and accurate, its computational cost is high for large-scale or high-frequency systems due to stringent stability conditions (the Courant-Friedrichs-Lewy condition) requiring very small time steps.

- **Lumped Element Approximation:** Representing a TL as a cascade of RLC sections (a ladder network) transforms the distributed problem into a set of ordinary differential equations (ODEs) solvable by efficient, adaptive ODE solvers such as MATLAB's ode23 **[19]**. This method offers a favourable balance of accuracy and speed but suffers from discretisation errors and, similar to FDTD, inherently assumes time-invariant parameters.

The Critical Research Gap:

While these advanced techniques (AWE, VF, NILT, Fast Convolution) have proven successful in microwave and high-speed digital applications, their application to the THz regime is nascent. A comprehensive, rigorous comparative analysis of these modern modelling techniques specifically for TLs with pronounced fractional-order, frequency-dependent behaviour is entirely lacking. There is no clear guidance on which method, or hybrid combination thereof, offers the optimal balance of accuracy, numerical stability, and computational efficiency for THz component design. My project will directly address this gap, moving beyond incremental improvements to provide a foundational analysis and deliver an efficient, validated modelling framework for the THz community.

Section Three: Proposed Research Methodology

This research will employ a computational methodology based on numerical simulation, algorithm development, and comparative analysis, primarily using software packages for accurate results such as MATLAB or Spice. The workflow is designed to be systematic, validation-centric, and achievable within a 3-4 year timeframe.

Year 1: Foundation and Benchmarking (Months 1-12)

- **Months 1-6:** Comprehensive analysis of the theoretical foundations of the key algorithms (AWE, VF/MVF, NILT, Piecewise-Linear Convolution). Implementation of a robust benchmark suite in MATLAB. This will include a reference NILT solver (based on principles from ([3] [4]) to generate "ground truth" data from analytical s-domain TL models incorporating fractional-order $R(\omega)$ and $G(\omega)$.
- **Months 7-12:** Rigorous benchmarking of established methods (FDTD, RLC ladders) against the NILT benchmark for standard test cases (step, pulse inputs). This phase will quantitatively reconfirm their limitations at THz frequencies and establish baseline performance metrics (RMSE, absolute error, CPU time). **Deliverable:** A technical report or conference paper (e.g., IEEE MTT-S International Conference on Numerical Electromagnetic and Multiphysics Modelling and Optimization - NEMO) establishing the benchmark framework.

Year 2: Core Algorithm Development and Hybridisation (Months 13-24)

- **Months 13-18:** Development and implementation of advanced modelling modules. This includes:

1. A numerically stabilized AWE variant, incorporating strategies like enforced conjugate symmetry and adaptive expansion point selection inspired by Complex Frequency Hopping.
2. Implementation and potential modification of VF/MVF algorithms (e.g., rationalfit in MATLAB).
3. Implementation of a fast convolution solver based on the PWL-SFC principles outlined in [15].

- **Months 19-24:** Exploration of hybrid approaches. A key innovation will be testing combinations, such as using VF to generate a stable initial pole-set for AWE, or integrating a PWL-SFC solver with a frequency-domain parameter extractor. **Deliverable:** A journal paper (e.g., *IEEE Transactions on Microwave Theory and Techniques*) on the developed hybrid algorithm and its initial performance.

Year 3: Validation, Testing, and Dissemination (Months 25-36)

- **Months 25-30:** Extensive validation of the developed models against the NILT benchmark across a wide range of scenarios: different TL geometries (microstrip, stripline), varying degrees of frequency-dependent loss, and complex modulated input signals.
- **Months 31-36:** Comparison against commercial software packages (e.g., Keysight ADS, Cadence) where possible. Exploration of collaboration for measured validation**:** We will actively seek collaboration with industrial partners (e.g., semiconductor foundries, antenna manufacturers) or internal DCU research groups with THz measurement capabilities to validate models against empirical data. **Deliverable:** A high-impact journal paper (e.g., *IEEE Transactions on Circuits and Systems I: Regular Papers, COMPEL*) presenting the comprehensive comparative study and validation results. Active participation in a major conference (e.g., IEEE MTT-S International Microwave Symposium (IMS), European Microwave Conference (EuMC)) to present findings.

Year 4: Finalisation, Thesis Writing, and Technology Transfer (Months 37-48)

- **Months 37-42:** Final refinements to the modelling framework. Packaging of the most successful algorithms into a user-friendly toolbox or library. Preparation of the final journal paper from the thesis core contributions.
- **Months 43-48:** Completion of thesis writing and submission. Dissemination of the open-source toolbox and dataset to the academic and industrial community. **Deliverable:** PhD Thesis. Final journal submission (targeting a high-profile journal such as *Proceedings of the IEEE* or *Nature Electronics or IEEE Transactions on Electromagnetic Compatibility or COMPEL*). Presentation of final results at a premier conference (e.g., International Conference on Infrared, Millimeter, and Terahertz Waves - IRMMW-THz).

Validation and Comparison Strategy:

The superiority of the proposed models will be demonstrated through:

1. **Quantitative Metrics:** RMSE, absolute error, and computational time compared to the NILT solution.
2. **Benchmarking against Literature:** Comparison of results for standard TL structures against published data in reputable journals.
3. **Software Comparison:** Where feasible, comparison of key performance indicators (e.g., S-parameters, eye diagrams) against simulations from established commercial tools.
4. **Collaborative Measurement:** Pursuing partnerships to obtain measured S-parameters of fabricated THz structures for ultimate validation.

Section Four: Justification for Proposed Research

This research is not merely an academic exercise; it is a critical enabler for future technological advancement, with significant contributions to knowledge, industry, and society.

Contributions to Academic Knowledge:

This PhD will make several novel contributions to the body of knowledge:

1. Theoretical Contribution: This PhD will provide a comprehensive comparative analysis of modern modelling techniques (FDTD, RLC, AWE, VF/MVF, and Fast Convolution) in the context of THz transmission lines with pronounced fractional-order behaviour. Crucially, it will investigate the novel hybridization of AWE with complementary methods such as fast convolution to overcome its inherent numerical limitations when applied to wideband THz systems. The findings will define the specific strengths, weaknesses, and optimal application domains for each method and their combinations.

2. Methodological Contribution: The development of numerically stable, wideband variants of these algorithms and the exploration of a novel hybrid framework represent a significant advance in computational electromagnetics. The strategies for handling strong frequency-dependence in time-domain simulation will be a valuable reference.

3. Empirical Contribution: The research will generate a valuable open-source dataset of exact and simulated THz TL responses under various conditions, serving as a benchmark for future algorithm development.

Contributions to Practice and Industry:

The practical implications are substantial:

1. Enabling 6G and Beyond: By providing accurate and efficient simulation tools, this research will directly accelerate the design cycle for THz components (antennas, filters, interconnects) essential for 6G networks, high-speed data centres, and sensing systems. designers can optimise performance and mitigate signal integrity issues before costly fabrication.

2. Efficient Design Automation: The developed compact models (pole-residue sets or convolution kernels) are compatible with and can be imported into commercial circuit simulators (e.g., Cadence, ADS). This enables system-level simulation of THz circuits that are currently impossible due to computational constraints.

3. Bridging a Critical Gap: This work addresses the critical gap between the theoretical understanding of THz wave propagation and the practical tools available to engineers to design with it. It moves beyond documenting the limitations of existing methods to providing a viable, superior solution.

4. **Economic and Societal Benefits:** The companies that will benefit most from this work include:

    - Telecommunications Equipment Manufacturers: (e.g., Nokia, Ericsson, Huawei) developing 6G infrastructure.
    - Semiconductor Companies: (e.g., Intel, GlobalFoundries, TSMC) designing THz integrated circuits.
    - EDA Software Vendors: (e.g., Cadence, Keysight, Synopsys) seeking to enhance their tool capabilities for the THz market.
    - Sensors and Imaging Companies: developing medical imaging and security scanners.
    - A dedicated section for engaging with these potential partners will be developed during the project to ensure practical relevance and technology transfer.

Engagement with Industry Partners:
To ensure the practical relevance and impact of this research, active engagement with industrial partners will be pursued throughout the project. This collaboration is crucial for grounding the theoretical work in real-world design challenges and facilitating technology transfer. We have initiated preliminary discussions with industry leaders to identify specific needs.

For instance, Klas Ltd., a company specializing in advanced networking and communications solutions, has expressed a strong interest in the potential applications of this research. Enda Sullivan, Chief Engineer at Klas Ltd., has stated that “The challenges of signal integrity at high frequencies are a major bottleneck in our pursuit of higher data rates and more compact systems. The limitations of current modelling tools become apparent as we push towards 100 GHz applications, where what we often call a '100 GHz' link is fundamentally built from multiple aggregated 25 GHz channels. A robust and computationally efficient modelling framework, as proposed in this PhD, could significantly improve the accuracy of our transmission line modelling. This would directly benefit the design of radio frequency front ends, high-density PCBs, and even next-generation Ethernet cables, enabling us to design more reliable and higher-performance systems”. This feedback from Klas Ltd. underscores the direct industrial need for this research. It highlights that the benefits extend beyond futuristic 6G systems to immediate challenges in existing high-frequency design

Feasibility and Commitment:
This project is accurately planned for a 4-year duration, with clear, achievable milestones. The required core software resource, an academic license for MATLAB, is available at DCU and is fundamental for the algorithm development and benchmarking central to this work. Furthermore, possible access to a high-performance computing (HPC will be explored for the most computationally intensive simulations (e.g., large-scale FDTD benchmarks or complex hybrid model evaluations). I am fully committed to disseminating the findings through at least four publications in top-tier IEEE journals (e.g., *IEEE Trans. on MTT*, *IEEE Trans. on CAS-I*, *International Journal of Circuit Theory and Applications,*) and through active participation in leading international conferences (e.g., IMS, EuMC, ECTC, ISCAS) to engage with the global research and industrial community. This work is a necessary and timely step to unlock the potential of the terahertz spectrum, offering a compelling return on investment for both academic and industrial stakeholders.

In conclusion, this PhD project is not merely an incremental improvement but a necessary step to unlock the potential of the terahertz spectrum. The demonstrated efficiency of advanced methods such as the θ-method and the clear industrial need from different tech companies underscore the timeliness and relevance of this research By creating computation efficient modelling frameworks, it will provide the foundational tools needed to design the next generation of wireless technology, offering significant contributions to theory, methodology, and industrial practice.

Section Five: Preliminary Bibliography

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